

EduGenie OS

AI-Powered Course Creation & Launch Platform

Product Requirements Document

Version v1.0 — Initial Draft	Date May 2026
MVP Scope 18 features — 7 AI Pipeline · 6 Creator · 3 Learner · 2 Commerce	Status Draft — For Review

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TL;DR (60-second read)

Field	Value
What	EduGenie OS — AI Course Creation & Launch Platform. Automated pipeline (Research → Curriculum → Scripts → Video → Quizzes → Roadmap → Sales Page) built on seven AI agents.
MVP scope	18 features: 7 AI Pipeline · 6 Creator Tools · 3 Learner Features · 2 Commerce. Full 10-module course built in < 4 hours.
Target customer	"Aarav the Knowledge Creator" — expert, educator, or L&D team who wants to sell their knowledge without production overhead.
Differentiator	Only platform that takes one topic brief → complete, deployed, selling course — content + video + quizzes + roadmap + sales page + payment — no other tool required.
Pricing	Starter free · Creator \$45/mo · Studio \$109/mo · Enterprise custom
Frontend	Next.js 14 + TypeScript + TailwindCSS (Creator OS + LearnSpace)
Mobile	React Native + Expo — learner course player + creator approvals (Phase 2)
Backend	Python 3.11 + FastAPI + Pydantic v2
Database	PostgreSQL 16 (Supabase) + pgvector + Redis
AI Pipeline	LangGraph supervisor · 7 agents · Anthropic Claude (primary) + GPT-4o (research) + Gemini (fallback)
Voice + Video	ElevenLabs (voice clone + TTS) · FFmpeg + AWS Batch (video render) · Whisper (captions)
Slides	Python-pptx + DALL-E 3 / Ideogram
Storage	AWS S3 + CloudFront CDN
Payments	Stripe (subscriptions + creator payouts via Connect)
Search	Algolia (marketplace)
Email	Resend
Observability	Langfuse (AI traces) + Sentry + Datadog APM
IaC	Terraform · AWS ECS Fargate · Vercel (frontend)
Timeline	Month 1–2: Foundation → Month 3–4: Alpha → Month 5–6: Public Beta → Month 9+: Enterprise

1. Executive Summary

Product Name: EduGenie OS — AI-Powered Course Creation & Launch Platform

One-Line Vision: An AI operating system that takes any topic brief and autonomously builds, packages, launches, and continuously improves a full online course product — content, video, quizzes, roadmap, and sales page — in a single session.

Target Users: Independent creators (coaches, consultants, educators), corporate L&D teams, and digital agencies who want to turn expertise into a sellable course product without production overhead.

High-Level Solution: A seven-agent AI pipeline (Intelligence → Architect → Scriptwriter → MediaForge → Evaluator → Launchpad → Optimizer) that runs sequentially with creator review gates between stages. Each agent is specialized: research, curriculum design, content writing, video production, assessment creation, marketing, and post-launch analytics. The creator reviews and approves each stage; the AI handles all production.

Expected Business Impact:

- Reduce course creation time from 150–300 hours to < 4 hours
- Enable creators with zero production skills to launch professional courses
- Target \$6M ARR by Month 18 via subscriptions + marketplace commission
- Creator median: first sale within 48 hours of publish
- Student completion rate target: > 45% (vs. industry average of 13–15%)

2. Problem Statement

Problem: Expert knowledge is trapped. Consultants, coaches, and educators with real expertise spend 150–300 hours producing a single course — writing, recording, editing, designing, and launching — before earning a dollar. Most give up. The creator economy's fastest growing segment (e-learning, \$400B+ by 2027) is underserved by tools that assist but never automate.

Who is affected

- Independent creators — coaches, consultants, academics — with expertise but no production capacity
- Corporate L&D teams — backlogged with training content requests, outsourcing at \$10K–\$50K per course
- Course agencies and studios — high manual production cost, low output margin
- Subject matter experts who want passive income but have no time for course production

Current solutions and their limits

Solution	Why it fails
Teachable / Thinkific / Kajabi	Hosting only. Creator still does 100% of production work — scripting, recording, editing, design, marketing copy.

Solution	Why it fails
AI writing tools (Jasper, ChatGPT)	Text only. Can't design curriculum, produce video, build assessments, or create a sales page. No end-to-end integration.
Video tools (Camtasia, Descript)	Reduce editing time but require recording. Still 40–80 hours of creator work per course.
Course creation agencies	Cost \$15K–\$80K per course, 8–16 week turnaround. Not accessible to 99% of creators.
DIY AI stacks (ChatGPT + ElevenLabs + Canva)	Fragmented, inconsistent, no deployment, no marketplace, no analytics — creator still owns all integration work.

Why now (May 2026)

- Foundation models (Claude Opus 4, GPT-4o) cross the quality threshold for production-grade instructional content
- Voice synthesis (ElevenLabs) produces learner-acceptable narration — no recording studio needed
- AI video generation (FFmpeg + ElevenLabs) is mature enough for professional slide-narrated video
- No competitor owns the full-stack AI course OS category — the category-creation window is open
- Post-pandemic e-learning adoption is permanent — learner market 3x larger than 2019

3. Target Users and Personas

Primary Customer (Wedge)

"Aarav the Knowledge Creator" — independent consultant or coach, 8–20 years of domain expertise, age 30–52, India/US/UK market.

- Pain: Has tried recording a course twice. Both times abandoned after 20 hours of work. Doesn't have editing skills, hates the camera, doesn't know how to write a sales page.
- Buys: EduGenie Creator tier (\$45/mo) and publishes 2–4 courses per year
- Wants: Full course done in one day, in his own voice, that actually sells
- Where to find: LinkedIn creator communities, coach/consultant Facebook groups, Substack creator economy newsletters

Secondary Customers (Phase 2 expansion)

Persona	Pain	Pricing tier
Corporate L&D Manager	Training content backlog 12–18 months, outsourcing budget exhausted, needs 20+ courses/year	\$109–Enterprise/mo (Studio)
Course Agency / Studio	Manual production doesn't scale, quality inconsistent, can't serve 10+ clients simultaneously	\$109/mo Studio or white-label
Academic Educator	Wants to reach beyond institution, zero marketing + production experience	\$45/mo Creator

Persona	Pain	Pricing tier
Creator Entrepreneur	Has an audience but no educational product — wants to productize fast	\$45/mo Creator
Education Startup	Needs a full course library fast without hiring content team	Enterprise / API access

Expected Usage Patterns

- Day 1: Sign up → submit topic brief → review market research → approve curriculum → first course draft starts
- Week 1: Full 10-module course reviewed, edited, published, first sales page live
- Month 1: 1–3 published courses, voice model trained, affiliate program active
- Month 3+: Regular course updates via Optimizer reports, bulk batches for L&D users, multi-language in use

4. Product Goals and Success Metrics

Business Goals

Metric	3 months	6 months	12 months
Paying creators	500	3,000	15,000
MRR	\$20K	\$200K	\$900K
Courses published	2,000	20,000	100,000
GMV (course sales)	\$200K	\$3M	\$15M
Enterprise accounts	2	10	30

Product Goals

- Activation: First course draft started within 30 minutes of signup (driven by 3-field topic brief)
- Creation speed: Full 10-module course (content + video + quizzes + sales page) complete in < 4 hours
- Retention: 65% of creators who publish one course publish a second within 60 days
- Time-to-revenue: Median creator earns first sale within 48 hours of publish
- Learner outcomes: Platform-wide course completion rate > 45%

AI-Specific KPIs

Metric	Target / Measurement
Curriculum coherence (SME audit)	> 88% rated 'logically sequenced' — monthly 5% sample audit
Content factual accuracy	< 2.5% error rate — monthly domain expert audit on random sample
Hallucination rate	< 2% — flagged [VERIFY] inline; creator review required before publish

Metric	Target / Measurement
Voice clone naturalness	> 4.0/5.0 — blind listening test at onboarding + monthly
Video generation latency	< 7 min per 10-minute lesson — AWS Batch FFmpeg p95
Quiz validity (instructional review)	> 85% rated 'valid and educationally sound' — bi-weekly audit
AI cost per full course build	< \$12 blended (Claude Opus + ElevenLabs + DALL-E)
Pipeline success rate	> 98% of builds complete without human-intervention error
Sales page conversion rate	> 3.5% from cold traffic — tracked per course via PostHog

5. Scope Definition

IN SCOPE — MVP (ships by Month 5)

18 features total: 7 AI Pipeline · 6 Creator Tools · 3 Learner Features · 2 Commerce features.

Seven AI pipeline stages (fully automated with review gates):

- Intelligence Agent — market research, competitor analysis, demand scoring, unique angle recommendation
- Architect Agent — full curriculum design (Bloom's taxonomy aligned), module/lesson structure, prerequisites
- Scriptwriter Agent — complete lesson scripts (2,000–4,000 words), examples, code samples, takeaways
- MediaForge Agent — slide decks (PPTX + PNG), ElevenLabs narration, FFmpeg video render, Whisper captions
- Evaluator Agent — module quizzes (MCQ + T/F + fill-blank), capstone project, scoring rubric, flashcards
- Launchpad Agent — sales page copy, pricing recommendation, 6-email launch sequence, 15 social posts
- Optimizer Agent — post-launch: drop-off analysis, quiz failure patterns, improvement report (weekly)

Creator tools:

- Creator Studio — full review + edit interface for every pipeline stage
- Voice Studio — voice model training from 10–30 min audio upload, test + deploy
- Slide editor — in-browser slide editing, layout swap, regenerate individual slides
- Landing page editor — WYSIWYG editing of AI-generated sales page, headline A/B testing
- Revenue dashboard — earnings by course, payout history, refund tracking, per-channel attribution
- Analytics dashboard — real-time student progress, completion heatmaps, quiz performance, AI weekly insight

Learner features:

- Course player — video, captions, speed control, chapter markers, note-taking, progress tracking
- Assessment interface — quiz, immediate feedback, score, retry, capstone submission

- Certificate system — auto-generated PDF + PNG, LinkedIn deeplink, public verification URL

Commerce:

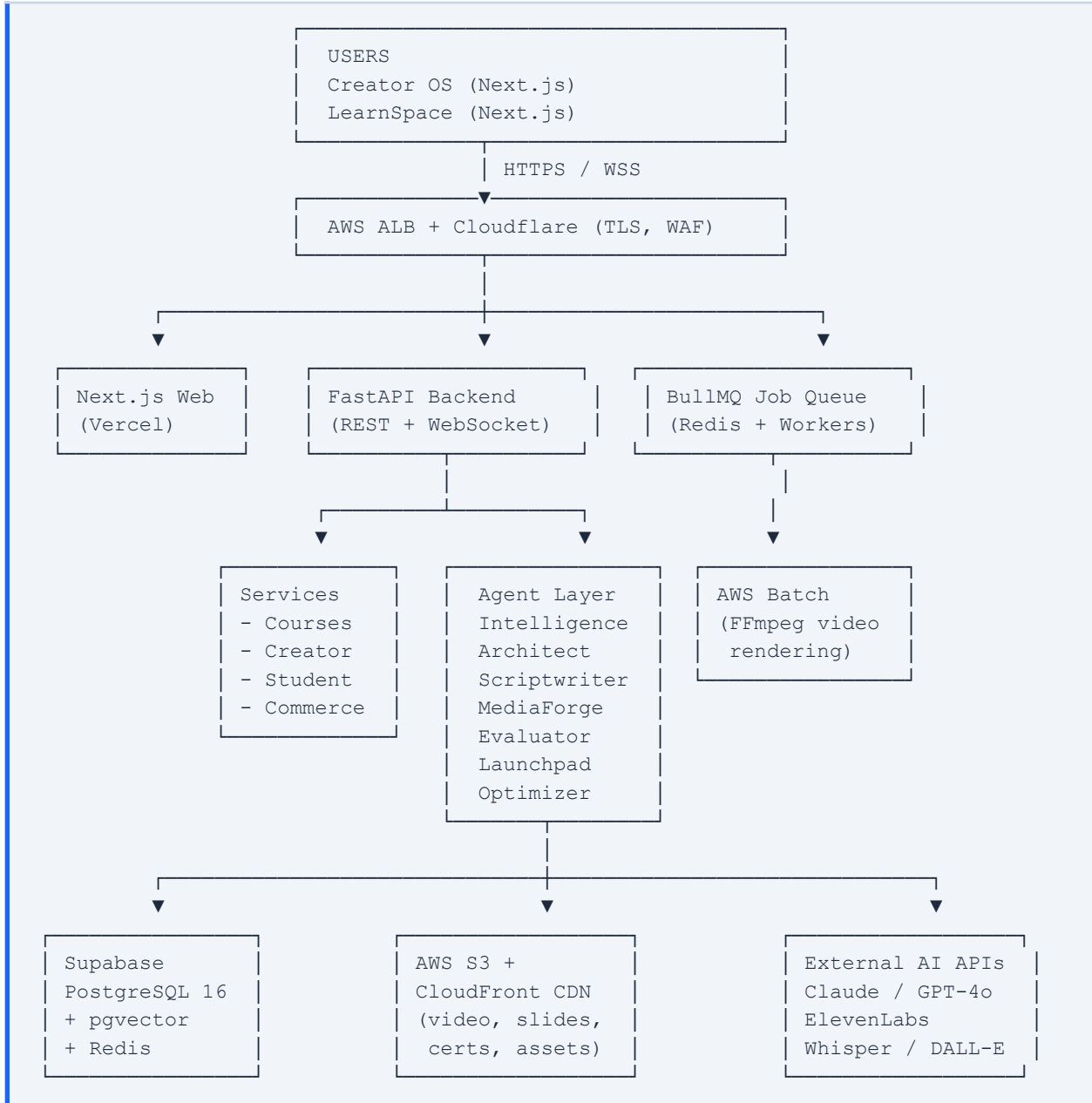
- Marketplace — Algolia search, filters, AI recommendations, course detail pages, free preview
- Payments + payouts — Stripe checkout, subscriptions, creator payouts (bi-weekly via Connect)

OUT OF SCOPE — Phase 1

Excluded	Target phase
Native iOS / Android learner app	Phase 2 — Month 8
Live cohort / synchronous classes	Phase 2 — Month 10
SCORM / xAPI LMS export	Enterprise — Month 12
Multi-instructor collaboration	Phase 2 — Month 9
Community / social features	Phase 2 — Month 7
Blockchain credential verification	Phase 3 — Year 2
White-label platform resale	Enterprise — Month 15
AI 1:1 tutoring sessions	Phase 2 — Month 11
Audio-only / podcast course format	Phase 2 — Month 9

6. System Overview (High-Level Architecture)

Architecture Diagram (ASCII)



Layer Breakdown

Layer	Technology + Responsibility
Frontend Web	Next.js 14 + TypeScript + TailwindCSS — Creator OS (Studio + Analytics + Revenue) + LearnSpace (player + quizzes + certs)
Mobile (Phase 2)	React Native + Expo — learner course player + creator approval notifications
API Gateway	AWS ALB + Cloudflare — TLS 1.3, rate limiting, WAF, routing

Layer	Technology + Responsibility
Backend	Python 3.11 + FastAPI + Pydantic v2 — REST + WebSocket APIs, async I/O
Agent Orchestration	LangGraph supervisor pattern — 7 agents, stateful pipeline, checkpointing, parallel execution
Job Queue	Redis + BullMQ — async course build jobs, video render dispatch, scheduled optimizer runs
LLM Routing	Claude Opus 4 (curriculum + scripts + copy) · GPT-4o (research + vision) · Haiku / Flash (quizzes + short tasks)
Voice + Audio	ElevenLabs API — voice model training, TTS narration, multi-language voices
Video Rendering	FFmpeg on AWS Batch — PNG frames + MP3 → 1080p MP4; Whisper → SRT captions
Slides	python-pptx + DALL-E 3 / Ideogram — structured slide generation, PPTX + PDF + PNG export
Database	Supabase PostgreSQL 16 + pgvector + RLS — multi-tenant data, vector search, auth
Cache + Queue	Redis — sessions, rate limiting, job queue, pipeline checkpoint state
Storage	AWS S3 + CloudFront — video files, slides, audio, certificates; signed URLs
Marketplace Search	Algolia — course catalog, faceted filters, personalized ranking, autocomplete
Payments	Stripe — subscriptions, Stripe Connect (creator payouts), Billing Portal, Radar (fraud)
Email	Resend — transactional + creator announcements + launch sequences
Observability	Langfuse (AI traces) + Sentry (errors) + Datadog APM + PostHog (product analytics)
IaC + Deploy	Terraform + AWS ECS Fargate (backend) + Vercel (frontend) + GitHub Actions CI/CD

Data Flow — Course Build (End-to-End)

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Creator submits topic brief (topic + audience + depth + tone + language)
↓
Intelligence Agent → web search + competitor scrape + keyword API → market report
↓ [REVIEW GATE: creator approves topic angle]
Architect Agent → curriculum JSON (modules, lessons, objectives, durations)
↓ [REVIEW GATE: creator reorders / approves outline]
Scriptwriter Agent → parallel script writing (N modules, fan-out) → lesson scripts (S3)
↓
MediaForge Agent → parallel per module:
  Slide Agent → slide JSON → python-pptx render → PPTX + PDF + PNG frames (S3)
  Voice Agent → ElevenLabs TTS → MP3 narration (S3, cached by hash)
  Video Agent → FFmpeg (PNG frames + MP3) → 1080p MP4 + Whisper SRT (S3)
↓

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Evaluator Agent → quiz JSON (questions, answers, explanations) + capstone brief + flashcards
↓
Launchpad Agent → sales page HTML + pricing rec + 6-email sequence + 15 social posts
↓ [FINAL REVIEW: creator reviews all sections → edits → approves]
Publish → DB record + S3 URLs locked + Algolia indexed + Stripe product created → live URL
↓ (post-launch, weekly)
Optimizer Agent → student analytics → improvement report → creator action items

Three Critical User Flows

Flow 1 — New Creator, First Course (< 4 hours total)

Sign up → Voice Studio (10-min audio upload) → voice model trained (25 min)
→ Topic brief submitted (2 min) → Intelligence Agent runs (12 min)
→ Creator reviews market report, picks angle → Architect designs curriculum (5 min)
→ Creator reviews curriculum, reorders 2 modules → Scriptwriter runs (parallel, 90 min)
→ MediaForge runs (parallel video render, 90 min) → Evaluator runs (20 min)
→ Launchpad runs (8 min) → Creator reviews all sections, edits 3 scripts
→ Publish → live URL with sales page + Stripe checkout → first social post

Flow 2 — Student Enrols and Completes Course

Student discovers course via marketplace search → views course detail page
→ Watches 2 free preview lessons → purchases via Stripe checkout
→ Enrols → starts Module 1 → watches video + takes notes
→ Completes module quiz (scores 85%) → progresses through all modules
→ Submits capstone project → creator reviews → marks complete
→ Certificate auto-generated → student shares on LinkedIn

Flow 3 — L&D Manager, Batch of 10 Courses

Enterprise admin uploads CSV (10 rows: topic, audience, depth, SME email)
→ System estimates cost + time → admin confirms
→ 10 pipeline jobs run in parallel (up to 30 on Enterprise)
→ Each SME receives review link → approves curriculum + content
→ Admin approves all → 10 courses published to org-private catalog
→ Employees assigned via HR system integration → completion tracked

7. Agentic AI Design ★ CRITICAL

Agent Definitions

Agent	Role	Capabilities	Tools + Approval boundary
Intelligence Agent	Market research and validation before any content is generated	web_search, competitor scrape (Apify), keyword demand (DataForSEO), learner question mining (Reddit/Quora), gap identification, confidence scoring	Read-only web + APIs. No approval needed. Creator reviews output before next stage.
Architect Agent	Full curriculum design using instructional design frameworks	Bloom's taxonomy alignment, module dependency mapping, prerequisite detection, duration estimation, learning objective writing	Read-only. Output reviewed at creator gate before Scriptwriter runs.
Scriptwriter Agent	Writes all lesson content — scripts, examples, code, case studies	write_lesson_script(), generate_code_examples(), generate_case_studies(), flag_uncertain_claims() [VERIFY tag]	Write to S3 only. Creator reviews and can edit or regenerate any script.
MediaForge Agent	Full multimedia production — slides, voice, video	generate_slide_json(), render_pptx(), synthesize_narration() [ElevenLabs], render_video() [FFmpeg], generate_captions() [Whisper], create_thumbnail()	Write to S3. Creator previews and can re-render any lesson. Approval before video publish.
Evaluator Agent	All assessment and engagement materials	generate_quiz_questions(), calibrate_difficulty(), write_answer_explanations(), design_capstone_project(), generate_flashcards()	Write to DB. Creator reviews quizzes; can edit, add, or remove any question.
Launchpad Agent	Marketing and commercialization — sales page, pricing, launch assets	write_sales_page(), suggest_pricing(), write_launch_emails(), generate_social_posts(), create_stripe_product(), write_seo_meta()	Write sales page + emails to DB; Stripe product created only on creator approval.
Optimizer Agent	Post-launch continuous improvement via student behavioral analysis	analyze_completion_dropoff(), identify_quiz_failure_patterns(), mine_discussion_questions(), generate_improvement_report(), flag_stale_content()	Read-only analytics. Generates recommendations; creator decides which to act on.

Memory Design

Short-term (per pipeline run, in Redis):

- Full LangGraph checkpoint state — all agent outputs accumulated in shared pipeline context object
- Creator review notes and regeneration instructions appended to context for targeted rewrites
- Intermediate agent outputs cached in Redis (5-minute TTL) for fast inter-stage transitions

- Pipeline is fully recoverable — any stage failure resumes from last checkpoint, not from start

Long-term (pgvector + PostgreSQL):

- Creator profile: tone/style preferences extracted from past editorial choices; expertise depth profile per domain
- Platform knowledge: research results, curriculum patterns from high-completion courses, content embeddings by topic
- Learner confusion signals: unresolved discussion questions indexed by concept — flagged in future content generation
- Voice model quality scores and phoneme corrections per creator — applied automatically on future narration

Reasoning Flow

Decision type	Pattern + Why
Intelligence designing research plan	Planning — full research strategy upfront before any web calls; cheaper to plan once
Scriptwriter writing lesson content	Reactive per section (hook → explanation → examples → takeaways) — predictable structure, easier quality gate
Optimizer diagnosing drop-off	Reflection / Reflexion — analyze behavioral trace → hypothesize cause → cross-reference quiz data → recommend fix
Orchestrator sequencing pipeline	State machine (deterministic stage transitions) — predictable, debuggable, easy to resume
MediaForge producing video	Parallel fan-out per module — all video renders dispatched simultaneously to AWS Batch

Guardrails (5 Layers)

- Policy guardrails: Topic screened at intake (content policy: hate, misinformation, fraud, dangerous DIY). Hard reject before any pipeline resources consumed.
- Content accuracy guardrails: Scriptwriter uses hedged language for uncertain claims; all uncertain assertions marked [VERIFY]; creator must explicitly dismiss each flag before video generation.
- Regulated domain guardrails: Medical / legal / financial topics trigger mandatory disclaimer injection into lesson content and sales page. AI cannot generate content constituting professional advice.
- Voice + IP guardrails: Consent confirmation before voice model training; ElevenLabs public-figure voice similarity detection; voice model locked to creator account, not transferable.
- Output guardrails: Originality.ai scan on all scripts (similarity > 12% triggers regeneration); quiz validation (every question has exactly one unambiguous correct answer); PII scan on creator-uploaded reference content via Presidio.

Autonomy Configuration (per pipeline stage)

Action	Default behavior
Web search + competitor scrape	Auto — read-only research, no creator approval needed

Action	Default behavior
Curriculum JSON generation	Auto — presented to creator at review gate before proceeding
Script writing to S3	Auto — presented to creator; [VERIFY] flags highlighted
Slide render + video render to S3	Auto — preview available; creator can re-render any lesson
Quiz + assessment generation	Auto — creator reviews + edits before publish; can add/remove questions
Sales page copy + email sequence	Auto — creator reviews + edits inline; publish requires explicit approval
Stripe product creation	Approval required — only executes after creator clicks 'Publish Course'
Course update (module regeneration)	Auto for script + slides; video render requires creator preview approval
Optimizer improvement recommendations	Auto report generated; all recommended changes require creator action

8. Feature Breakdown (Functional Requirements)

v1.0 scope: 18 features total. 7 AI Pipeline · 6 Creator Tools · 3 Learner Features · 2 Commerce. Each feature: name · description · user story · acceptance criteria · edge cases.

AI Pipeline Features (7 features)

F1 — AI Market Intelligence & Course Validation

Description: Intelligence Agent analyses the proposed topic before any content is written: market demand, competitor landscape, learner questions, content gaps, and pricing benchmarks. Delivers a validated strategic brief with unique angle options.

User story: As a creator, I want to know if my course topic has real demand, who my competitors are, what angle to take, and what to charge — before I invest any time building — so I can decide with data not gut feel.

Acceptance criteria: Market report delivered in < 12 min. Contains: demand score (0–100), 5+ competitor courses with module detail, top 30 learner questions (Reddit/Quora/LinkedIn), 3 unique angle options, price range recommendation, 'go confidence' score. Creator can select angle or request 3 alternatives. All data sourced from live search — no stale cache.

Edge cases: Topic is too broad (e.g. 'Marketing') → agent suggests 5 specific validated sub-niches with demand scores. Topic has zero demand → honest report with 3 pivot recommendations, not blocked. Topic is in regulated domain (medical, legal, financial) → domain sensitivity flag added; compliance disclaimer auto-injected downstream.

F2 — AI Curriculum Architect

Description: Architect Agent designs a complete, pedagogically sound course structure using Bloom's Taxonomy and backward design: modules, lessons, learning objectives, prerequisite map, and duration estimates.

User story: As a creator, I want a fully structured curriculum with clear learning objectives for every lesson and a logical progression from beginner to outcome — designed by AI so I don't need instructional design expertise.

Acceptance criteria: Curriculum generated in < 6 min. Output: 8–16 modules, 3–8 lessons per module, one Bloom's-aligned learning objective per lesson (cognitive level labeled), prerequisite dependency map, estimated video duration per lesson, total course time. Creator can drag-and-drop to reorder, add/remove modules, rename lessons. Changes trigger automatic dependency re-validation.

Edge cases: Creator wants very short course (3 modules) → architect reduces scope but maintains coherence; warns if depth is insufficient for promised outcome. Non-linear learning path (choose-your-path) → branching curriculum with conditional module unlocks. Creator reorders modules creating a pedagogical contradiction → flagged with suggested resolution.

F3 — AI Lesson Script Engine

Description: Scriptwriter Agent writes complete lesson scripts for every lesson in the approved curriculum.

Structure: attention hook → objective statement → concept explanation → 3+ examples → real-world application → common mistakes → key takeaways → transition to next lesson. Technical topics include runnable code examples with line-by-line annotation.

User story: As a creator, I want every lesson to have a complete, engaging script in my exact tone and depth — with hooks, examples, and smooth transitions — so students stay engaged and actually learn.

Acceptance criteria: All scripts for a 10-module course generated in < 2 hours (parallel). Each script: 2,000–4,000 words, structured sections marked, min 3 examples per concept, common mistakes addressed, key takeaways (5–7 bullets), transition hook. [VERIFY] tags on uncertain claims — creator must explicitly dismiss before video generation. Technical courses: 2–3 runnable code examples per lesson with syntax-highlighted block.

Edge cases: Creator wants micro-course scripts → configurable length (600–4,000 words). Script has factually uncertain claim → [VERIFY] tag with source recommendation; claim count shown in review UI. Creator pastes their own existing content → Scriptwriter restructures and enriches while preserving creator's original knowledge.

F4 — AI Slide Deck & Media Production

Description: MediaForge Agent converts each lesson script into a designed slide deck (PPTX + PDF + PNG frames), generates ElevenLabs voice-cloned narration (MP3), renders 1080p video (FFmpeg: slides + narration + captions), and produces a course thumbnail (DALL-E 3).

User story: As a creator, I want professional slide decks and narrated video lessons produced automatically from my scripts — in my own voice, with clean design — so my course looks like it was made by a production studio.

Acceptance criteria: Slide deck: 8–22 slides per 10-min lesson, PPTX + PDF + PNG; title/concept/diagram/code-block/takeaway slide types auto-selected. Video: 1080p MP4, stereo AAC, Whisper captions (> 95% accuracy), chapter markers, branded 5-sec intro/outro. Render time: < 7 min per 10-min lesson. Creator previews any 90-sec segment before approving full render. Individual segments re-renderable with revised script without re-generating entire lesson.

Edge cases: Voice clone training audio has background noise → noise suppression applied; quality score shown; re-upload prompted if below 3.8/5.0 threshold. Script has unusual technical terms → custom pronunciation dictionary editable per course before render. Creator uploads branded PPTX template → agent respects brand fonts, colors, logo position. Video render fails at FFmpeg → diagnostic logged; retry with adjusted encoding; creator notified with ETA.

F5 — AI Assessment & Evaluator Engine

Description: Evaluator Agent creates all assessment materials: module quizzes (MCQ, T/F, fill-blank, matching), difficulty calibration, answer explanations, a mid-course assignment, spaced repetition flashcard set, and a final capstone project with detailed scoring rubric.

User story: As a creator, I want assessments that genuinely test whether students learned — not just trivia quizzes — so my course produces real outcomes and earns strong reviews.

Acceptance criteria: Quizzes for all modules generated in < 30 min. Per module: 5–12 questions, mix of types, each with 4 options, correct answer, 2–3 sentence explanation, Bloom's level label, difficulty score. Capstone: project brief, 4–8 deliverables, timeline, 10-point rubric. Flashcard set: 20–30 cards per module. Creator edits, adds, or removes any question before publish.

Edge cases: Topic is subjective (design, leadership, creative writing) → reflective and portfolio-based assessments generated instead of factual MCQ. All students fail a quiz post-launch (< 30% pass rate) → Optimizer flags; Evaluator triggered to review and revise. Compliance course → stricter pass threshold (80%), attempt limit (3), certificate withheld until all passed.

F6 — AI Launchpad — Sales Page, Pricing & Launch Assets

Description: Launchpad Agent produces every asset needed to sell the course: landing page copy (headline variants, benefits, curriculum preview, FAQ, instructor bio, objection handling, CTAs), pricing recommendation, 6-email launch sequence, 15 social posts (LinkedIn / Twitter-X / Instagram), and SEO meta data.

User story: As a creator, I want a professional sales page and a full launch marketing kit generated automatically — so I can go live immediately without hiring a copywriter or spending days on page design.

Acceptance criteria: Sales page generated in < 6 min. Sections: 5 headline A/B variants, sub-headline, pain-point list, benefits list (10 bullets), full curriculum preview, instructor bio, 3-objection FAQ, 10–15 FAQ items, 2 testimonial placeholders, pricing section with value-stack framing, money-back guarantee. Deployed at [creator].edugenie.io/[slug]. Custom domain CNAME supported. Pricing recommendation includes: recommended price, floor, ceiling, early-bird %, payment plan structure, 3 revenue scenarios. Creator edits all sections inline via WYSIWYG editor.

Edge cases: Creator has no testimonials (new creator) → social proof replaced with 'Join first cohort' early-adopter framing. Technical audience → AI adjusts tone from emotional to ROI/analytical. Creator overrides pricing recommendation significantly → soft warning with projected revenue impact at chosen price. Creator wants multiple tiers (basic/premium) → pricing table with tier features auto-added to sales page.

F7 — AI Optimizer — Post-Launch Improvement Engine

Description: Optimizer Agent runs weekly post-launch analysis: student completion drop-off heatmap, quiz failure patterns, discussion question cluster mining, learner sentiment from reviews, competitive gap alerts (new competitor courses on same topic), and a prioritized improvement report with one-click regeneration links for top issues.

User story: As a creator, I want EduGenie to tell me every week exactly what to improve in my course — which lesson to rewrite, which quiz question is confusing students, which module has high drop-off — with evidence, so I improve quality continuously without being a data analyst.

Acceptance criteria: Improvement report generated weekly for courses with ≥ 10 enrolled students. Sections: Critical Issues (severity $\geq 8/10$), Content Gaps (concepts in discussion questions not covered in lesson), Drop-off Analysis (top 5 lessons by abandonment + diagnostic hypothesis per lesson), Assessment Issues (questions with > 55% failure rate), Pacing Problems (duration vs. actual), Sentiment Analysis (NLP of reviews + discussion). Each

issue has severity score, evidence, and 'Fix This' button → triggers targeted regeneration of only the affected section.

Edge cases: Course is brand new (< 10 students) → report deferred; student count milestone shown. All metrics excellent → positive benchmarking report with 'next course topic' suggestions based on audience interests.

Creator ignores reports for 6+ weeks → 'Revenue at Risk' notification showing estimated uplift if top recommendation implemented.

Creator Tool Features (6 features)

F8 — Creator Studio — Full Review & Edit Interface

Description: The central creator workspace where all pipeline outputs are presented stage by stage — with inline editing, regeneration-with-note, approval gates, and progress tracking through the full course build process.

User story: As a creator, I want to review, edit, and approve every AI-generated section before it goes to the next stage — with the ability to regenerate anything with a simple instruction — so I stay in control while the AI does the heavy lifting.

Acceptance criteria: Six-stage review flow: (1) Market Report → angle selection, (2) Curriculum → drag-drop reorder + inline edit, (3) Scripts → inline edit + [VERIFY] flag review + regeneration field, (4) Slides + Video → preview + segment regeneration, (5) Quizzes + Capstone → question editor, (6) Launch Assets → WYSIWYG sales page + email preview. Progress indicator (% of stages approved) always visible. Creator can return to any approved stage and re-open for changes. All edits version-controlled.

Edge cases: Creator closes browser mid-review → LangGraph checkpoint state preserved; resumes exactly where left off. Creator regenerates a script after the video is already rendered → video automatically marked 'stale' and re-render prompted. Creator approves curriculum then adds 2 new modules → Scriptwriter, MediaForge, and Evaluator re-run only for the new modules, not the full course.

F9 — Voice Studio — Voice Model Training & Management

Description: A dedicated interface for training, testing, managing, and deploying the creator's personal AI voice model from uploaded audio samples — enabling all video narrations to sound authentically like the creator.

User story: As a creator, I want all my course videos narrated in my own voice so my students feel a personal connection with their instructor — without me recording a single sentence.

Acceptance criteria: Voice model trained from 10–30 min uploaded audio (WAV/MP3) in < 25 min. Quality score shown: naturalness (0–5), clarity (0–5), expressiveness (0–5). Creator tests model with any custom text (unlimited). Speaking rate (0.5x–1.5x), pitch variation, pause timing configurable per course. Custom pronunciation dictionary: add terms, acronyms, names with phonetic spelling. Creator can train a second 'professional tone' variant. Voice model update: upload new samples → retrain → apply to new courses (existing videos unchanged unless creator triggers re-render).

Edge cases: Audio has significant room echo → noise reduction applied; quality score shown; re-upload recommended if below 3.8/5.0. Creator wants a team member's voice for a specific course → second voice model on Studio+ plans; consent confirmation required. Voice model produces mispronunciations on technical acronyms → pronunciation dictionary edit resolves; no full model retraining needed.

F10 — Analytics Dashboard — Student Intelligence & Learner Insights

Description: Real-time analytics dashboard showing every meaningful student metric: enrollment cohorts, lesson completion rates, video watch depth, quiz performance distributions, drop-off heatmaps, time-to-complete histograms, device/geo data, and a weekly AI insight narrative in plain English.

User story: As a creator, I want to see not just enrollment numbers but exactly where students get stuck, which lessons they love, which quizzes they're failing — and be told what to do about it in plain language.

Acceptance criteria: Dashboard updates in real-time (< 30s lag). Sections: Overview (enrolled, active this week, completion rate, avg quiz score, revenue); Lesson Analytics (per-lesson: watch depth %, replays, notes taken, discussion posts); Quiz Analytics (avg score, pass rate, most-failed question); Drop-off Heatmap (visual, above-average abandonment flagged); Cohort Analysis (completion rates by enrollment week). Weekly AI narrative: 2–3 paragraph plain-English explanation of the most important trend + one specific recommended action. Export: CSV (full data) + automated weekly PDF digest emailed to creator.

Edge cases: Course has < 10 students → 'Insufficient data' placeholder with milestone tracker. Creator has 10+ courses → cross-course comparison dashboard with per-course sparklines. All metrics excellent → benchmark comparison report + next-course topic recommendations based on audience interest signals.

F11 — Course Update & Version Management

Description: Allows creators to update any published course section — regenerating specific lessons, modules, quizzes, or the full course — with full version control, student change notifications, and support for maintaining multiple active course versions.

User story: As a creator, I want to update individual lessons when content becomes outdated or feedback reveals gaps — and have EduGenie regenerate only that section in minutes — so my course stays fresh without rebuilding from scratch.

Acceptance criteria: Creator selects any lesson/module from published course. Update modes: (1) Selective Regenerate — specific lesson with creator's notes, (2) Content Refresh — AI checks for topic freshness before rewriting, (3) Bulk Staleness Scan — AI flags highest-risk modules across all courses. Version control: each update creates Course v[X]. Creator can maintain 3 active versions. Existing students: email notification with changelog; updated lessons marked 'New' in player; completion progress preserved. Schedule updates for specific publish date/time.

Edge cases: Creator updates a foundational module referenced by later modules → downstream content dependency checker flags which later lessons may need corresponding updates. Course was in English → Update Engine supports adding translated module versions without regenerating original. Student is mid-lesson when update publishes → in-progress lesson not interrupted; update applies from next lesson onward.

F12 — Affiliate & Revenue System

Description: Full affiliate program management — unique referral link generation, real-time conversion tracking, automatic commission calculation, Stripe payouts to affiliates, fraud detection, and a creator revenue dashboard with earnings by course/channel/period.

User story: As a creator, I want to run an affiliate program and see all my revenue — by course, by channel, by affiliate — and have payouts handled automatically, so I can scale marketing and manage money without spreadsheets.

Acceptance criteria: Affiliate program: commission rate 10–70%, cookie window 7/14/30/60 days, auto or manual approval. Unique tracking links per affiliate. Real-time creator dashboard: clicks, conversions, revenue, commission owed per affiliate. Affiliate dashboard: link, click count, conversion count, pending earnings, payout history. Payouts: monthly rolling 30-day cycle via Stripe Connect. Revenue dashboard: total GMV, net revenue (after fees), by course, by channel (direct/marketplace/affiliate/promo), refund rate, MoM growth. Tax docs: 1099 (US), annual earnings summary.

Edge cases: Referred customer requests refund → commission reversed from affiliate pending balance; clawback within 60-day window. Affiliate generates high click volume with near-zero conversions → fraud detection flags account; creator notified. Creator sets commission making net payout negative after platform fee → warning with margin calculation shown; submission blocked if result is negative.

F13 — Multi-Language Course Generation

Description: Generates the entire course — scripts, slides, video narration, quizzes, and sales page — in up to 6 languages simultaneously, with cultural localization (not just word-for-word translation) and language-native AI voice models.

User story: As a creator, I want to launch my course in Spanish, Portuguese, and French automatically — so I can reach 3x more students without paying translation agencies.

Acceptance criteria: Creator selects target languages at creation or for existing course. Supported at launch: English, Spanish, Portuguese, French, German, Japanese. Full pipeline output translated and culturally localized per language in < 3 hrs/language (parallel). Cultural localization: examples adapted to local context, idioms replaced, currency/date formats adjusted. Translated video: language-appropriate ElevenLabs voice model (native accent). Translated sales page: localized URL (/es/, /pt/, /fr/) with hreflang SEO tags. Pricing adjusted per market (purchasing power parity suggestion).

Edge cases: Creator wants to translate only the sales page to test demand in a new market → partial localization mode supported. Translated content rated poorly by students in a specific market (Castilian vs. Latin American Spanish) → market-specific dialect option in language settings; re-localization available. Creator updates Module 3 in English → cascade update option: re-generate all translations of Module 3 only.

Learner Features (3 features)

F14 — Course Player & Progress Tracking

Description: A clean, distraction-free video course player with captions, speed control, chapter markers, synchronized note-taking, per-lesson progress tracking, and a discussion Q&A board with AI instant responses sourced from course content.

User story: As a student, I want to learn comfortably with a clean player that tracks my progress, lets me take notes, and gives me instant answers to questions — so I stay focused and feel supported throughout the course.

Acceptance criteria: Video player: speed control (0.5x–2x), captions toggle (editable by creator), fullscreen, chapter markers, auto-resume from last position. Note-taking: synchronized to lesson timestamp, exportable as PDF study guide. Progress: lesson checkmarks, module % complete, overall course progress bar, estimated time remaining. Discussion: per-lesson Q&A, AI responds in < 60 sec (sourced from lesson script + course content, labeled 'AI Response'), creator verified answers shown with green badge, upvoting, threading.

Edge cases: Student's internet connection drops mid-video → video resumes from exact pause point on reconnect. AI discussion response is wrong (factual error in source material) → creator can edit AI response + mark 'Creator Verified'; wrong response flagged for improvement report. Student posts question in a different language → AI detects language, responds in same language with note suggesting creator add translation.

F15 — Assessment Interface — Quizzes, Capstone & Certificates

Description: A full-featured assessment UI: progressive quiz delivery with immediate feedback, capstone project submission workflow, certificate auto-generation (PDF + PNG + LinkedIn share), and a public certificate verification page.

User story: As a student, I want to test my knowledge with engaging quizzes, submit a real-world capstone project, and earn a professional certificate I can share on LinkedIn — so completing the course has tangible career value.

Acceptance criteria: Quiz: one question at a time, progress bar, immediate answer feedback (correct/incorrect + explanation), retry after failing (different question order), score summary. Certificate: auto-generated when all modules completed and all quizzes passed (threshold configurable, default 70%). Certificate elements: student name, course title, creator name, completion date, unique 16-char verification ID, EduGenie verification URL, QR code. LinkedIn 'Add to Profile' deeplink pre-populated. Verification page: publicly accessible, shows Active/Revoked status. PDF + PNG download.

Edge cases: Student fails to meet pass threshold → certificate withheld; specific failing modules shown with retry prompts; progress toward certificate shown as %. Creator wants to revoke certificate for academic dishonesty → revocation flow: reason logged, certificate marked Revoked on verification page, student notified. Student requests name correction → correction flow: creator approves → new cert with same verification ID + 'Revised' timestamp on verification page.

F16 — Course Marketplace & Discovery

Description: Public-facing marketplace where all published EduGenie courses are searchable, filterable, and purchasable — with AI-powered personalized recommendations, curated topic collections, trending courses, and free lesson previews.

User story: As a student, I want to discover and buy courses quickly through smart search and recommendations — and feel confident purchasing based on course previews, honest ratings, and transparent creator signals.

Acceptance criteria: Algolia-powered search returns results in < 500ms. Filters: topic category (50+), price range, duration, difficulty, language, rating, 'New' / 'Updated'. Course card: title, thumbnail, creator name + avatar, star rating, review count, enrolled count, video hours, price. Free preview: first 2 lessons of any course before purchase. AI recommendations: 5 personalized on homepage based on browsing + enrollment history. Course detail page: full curriculum expandable preview, instructor bio, reviews, money-back badge, enroll CTA.

Edge cases: New marketplace with few courses → 'Editor's Picks' and 'Trending This Week' curated manually; quality bar enforced for listing. Search returns zero results → AI suggests related topics + 'notify when available' option. Student has no history (new user) → recommendations default to trending + top-rated in self-selected interest categories from onboarding.

Commerce Features (2 features)

F17 — Payments, Subscriptions & Creator Payouts

Description: Full Stripe-powered commerce layer: student one-time course purchases, creator SaaS subscriptions, creator payouts via Stripe Connect, VAT/tax handling, promo code engine, and refund processing.

User story: As a creator, I want my students to be able to purchase my courses easily in any currency, my earnings to be deposited automatically bi-weekly, and my subscription to give me access to the tools I need — all without touching a payment system manually.

Acceptance criteria: Student checkout: Stripe hosted payment page, 135+ currencies, 3DS authentication, Stripe Radar fraud protection. Course purchase confirmation email via Resend. Creator payouts: bi-weekly via Stripe Connect to creator's bank account; minimum \$25 payout threshold. Subscription: 4 tiers (Starter free, Creator \$45/mo, Studio \$109/mo, Enterprise custom) + annual discount. Promo codes: % or fixed, max uses, validity window, per-course or all-courses. Refund processing: full or partial refund from creator dashboard; automatic affiliate commission reversal within 60-day window.

Edge cases: Payment fails at checkout → Stripe retry logic; clear error message; no accidental double-charge. Creator hasn't connected bank account → earnings accumulate in wallet; visual reminder on every dashboard visit. Student in a country where Stripe isn't available → alternative payment provider roadmap (Razorpay for India, Phase 2). VAT required for EU transaction → Stripe Tax calculates and applies automatically.

F18 — Enterprise Batch Course Generation

Description: Allows L&D teams and agencies to submit a batch of 5–30 course topics via CSV upload, process all in parallel with shared org brand settings and SME review routing, and manage all jobs from a unified production Kanban dashboard.

User story: As an L&D manager, I want to submit our quarterly training plan (20 topics) and have EduGenie build all 20 courses in parallel with our branding and standards — reviewed by the relevant SME — so I can deliver a year's training content in two weeks.

Acceptance criteria: Admin uploads CSV (columns: topic, audience, depth, sme_email, due_date, compliance_flag). System validates, estimates total cost and time, requests confirmation. Parallel processing: up to 30 simultaneous jobs on Enterprise. All courses inherit org-level settings: brand template, logo, voice model, tone guidelines. SME routing: each SME receives unique review link per course (no EduGenie account required). Production dashboard: Kanban (Queued → Research → Architecture → Content → Media → Assessment → Launchpad → Review → Published), estimated completion time per course. Failed jobs flagged; other jobs continue; retry individual jobs without re-running batch.

Edge cases: One topic in batch triggers content policy violation → that job paused and flagged; all others continue. SME unresponsive for 48h past review deadline → automatic escalation email to admin; admin can approve on SME's behalf. Batch contains duplicate topics → system flags before starting; admin can merge or keep both. Organization requires multi-stage approval (SME → Legal → Admin) → configurable review workflow per org.

9. API and Integration Requirements

Internal APIs

Base: /api/v1/

Area	Key endpoints
Auth	POST /auth/{login,signup,refresh,magic-link}, GET /auth/me
Creators	GET/PATCH /creators/{id}, GET /creators/{id}/courses, GET /creators/{id}/revenue
Courses	GET/POST /courses, PATCH /courses/{id}, POST /courses/build (submit brief), GET /courses/{id}/pipeline (status), POST /courses/{id}/publish
Pipeline stages	POST /courses/{id}/stages/{stage}/approve, POST /courses/{id}/stages/{stage}/regenerate (+ instructions body)
Voice models	POST /voice/train (upload audio), GET /voice/models, POST /voice/test, DELETE /voice/{id}
Lessons	GET /courses/{id}/lessons, GET /lessons/{id}/script, GET /lessons/{id}/video, GET /lessons/{id}/slides
Quizzes	GET/PATCH /courses/{id}/quizzes, POST /quizzes/{id}/attempt, GET /quizzes/{id}/results
Students	GET /students/{id}, GET /students/{id}/enrollments, GET /students/{id}/progress
Enrollments	POST /enrollments (Stripe webhook triggers), GET /enrollments/{id}/progress, POST /enrollments/{id}/complete
Certificates	POST /certificates/generate, GET /certificates/verify/{code} (public)
Analytics	GET /analytics/courses/{id}/overview, GET /analytics/courses/{id}/lessons, GET /analytics/courses/{id}/quizzes, GET /analytics/courses/{id}/improvement-report
Marketplace	GET /marketplace/search (Algolia proxy), GET /marketplace/courses/{id}, GET /marketplace/recommendations
Affiliates	POST /affiliates, GET /affiliates/{id}/stats, GET /affiliates/{id}/payouts
Enterprise batch	POST /batch/jobs, GET /batch/jobs/{id}, POST /batch/jobs/{id}/retry
Webhooks	POST /webhooks/stripe (incoming), POST /webhooks/outbound (config)
Health	GET /health, GET /health/detailed

WebSocket:

- WS /ws/pipeline/{job_id}/live — real-time pipeline stage progress feed (stage name, % complete, ETA)
- WS /ws/analytics/{course_id} — live student enrollment + completion events

External APIs (Integrations)

Category	Integration	Notes
LLM (primary)	Anthropic Claude API — claude-opus-4 for curriculum + scripts + copy; claude-haiku for quizzes + short tasks	Model routing: Opus for depth; Haiku for cost; cost tracked per course_id
LLM (research)	OpenAI GPT-4o — Intelligence Agent (web browsing + vision for competitor screenshots)	GPT-4o with Tools; vision for competitor course page analysis
LLM (fallback)	Google Gemini Pro — secondary fallback when Anthropic + OpenAI unavailable	Activated via model routing config; not in primary path
Voice synthesis	ElevenLabs API — voice model training, TTS narration, multilingual voices	Creator plan; one model per creator; narration cached by (script_hash + voice_id)
Image generation	DALL-E 3 + Ideogram — slide illustrations, thumbnails, hero images	Ideogram preferred for text-in-image; DALL-E for illustrations
Captions	OpenAI Whisper (Audio API) — STT transcription of narration for SRT caption generation	Runs post-render; output: SRT synced to audio timestamps
Plagiarism check	Originality.ai — originality + AI-content similarity scoring on all generated scripts	Async post-generation; score shown in Studio; > 12% triggers regeneration
PII detection	Microsoft Presidio (self-hosted) — PII scan on creator-uploaded reference content	Before any user content enters LLM context
Payments	Stripe — payments, subscriptions, Connect (creator payouts), Billing Portal, Tax, Radar	Source of truth for all financial transactions; webhook-driven sync
Search	Algolia — marketplace search, faceted filters, personalized ranking, autocomplete	Course index synced on publish/update; < 500ms search latency SLA
Email	Resend — transactional email, launch sequences, affiliate notifications, weekly digests	React Email templates; SPF/DKIM per creator custom domain
Storage + CDN	AWS S3 + CloudFront — video, slides, audio, certificates; adaptive bitrate; signed URLs	Multi-region CDN; S3 versioning on all media assets
Product analytics	PostHog — creator + student behavior, funnel analysis, feature flags, A/B testing	No PII in events; anonymized user IDs; GDPR-compliant
AI observability	Langfuse — full prompt + response logs, token usage, cost per agent, latency per stage	Every LLM call traced; cost attributed to course_id + creator_id
Video render	AWS Batch + FFmpeg — elastic scale for MP4 rendering; no fixed capacity limits	Dispatched per lesson by BullMQ job queue
Research (web)	Google Custom Search API + Bing Search API — Intelligence Agent research	Combined for breadth; results deduplicated and relevance-ranked

Rate Limits

Tier	Limits
Starter (Free)	3 course builds/month · max 5 modules per course · 5 marketplace listings

Tier	Limits
Creator (\$45/mo)	Unlimited builds · unlimited modules · all integrations · 1 voice model · 6 languages · priority queue
Studio (\$109/mo)	Everything in Creator · 5 team seats · up to 8 parallel builds · white-label storefront
Enterprise	Unlimited everything · unlimited seats · up to 30 parallel builds · custom rate limits · API access
API (public)	Standard tier: 200 req/min · Enterprise: 2,000 req/min · all endpoints versioned at /api/v1/

10. Data and Storage Design

Data Sources

- Creator input: topic briefs, script edits, quiz edits, voice audio samples, reference documents
- AI pipeline outputs: scripts, slides, video files, quiz JSON, sales page HTML — all stored in S3 with DB metadata
- Student events: enrollment, video progress (per-second), quiz attempts, discussion posts, note content
- Commerce data: Stripe transaction events synced via webhook; Stripe as source of truth
- Market research: Intelligence Agent web search results cached in Pinecone by (topic_hash, date)

High-Level Database Schema (Supabase PostgreSQL)

Table	Purpose	Multi-tenant key
organizations	Top-level tenant (org account)	(root)
creators	Creator profile, plan, voice model ID, brand settings (JSONB)	organization_id
courses	Course metadata, status, version, price, Stripe product ID	organization_id
course_versions	Version history with changelog per publish	course_id
modules	Module definition, position, learning objective, Bloom's level	course_id
lessons	Lesson metadata + S3 URLs (script, video, slides, captions)	module_id
quizzes	Quiz definition, questions JSONB, pass threshold, attempts config	lesson_id / module_id
students	Student identity, Stripe customer ID, locale	(global)
enrollments	Student ↔ Course, version, enrolled_at, completed_at, cert_id	course_id
progress	Per-lesson: watch_depth_pct, time_spent_sec, completed_at	enrollment_id
quiz_attempts	Per attempt: score, answers JSONB, passed, duration, attempted_at	enrollment_id
sales	Stripe payment intent → course + enrollment mapping, channel, promo	course_id
affiliates	Affiliate link, commission, clicks, conversions, total earned	course_id
certificates	verification_code, PDF URL, badge JSON, issued_at, revoked_at	enrollment_id
discussions	Per-lesson: content, AI response, confidence, creator verified	lesson_id

Table	Purpose	Multi-tenant key
pipeline_runs	Stage, status, model, tokens, cost_usd, started_at, completed_at	course_id
improvement_rpts	Weekly Optimizer report JSON, priority score, viewed_at	course_id
audit_logs	Immutable append-only: all AI decisions, creator actions, system events	organization_id

Vector Database (pgvector)

- Embeddings: OpenAI text-embedding-3-small (1536d) for knowledge store + course content semantic search
- Use cases: research cache (Intelligence Agent), curriculum pattern matching, content similarity (plagiarism baseline), learner confusion signal clustering
- Hybrid search: 0.65 vector + 0.35 BM25 for research retrieval

Data Privacy

- Multi-tenant via organization_id + creator_id on every table; Supabase RLS policies enforce tenant isolation
- PII encrypted at rest (AES-256 application-level for student names/emails); TLS 1.3 in transit
- Creator content: creator owns all AI-generated content — EduGenie claims no IP rights or training rights
- Voice model audio: encrypted in S3; accessible only by creator's account and ElevenLabs integration
- GDPR: right to access (30-day export), right to deletion (90-day processing), data portability, DPA agreements with sub-processors
- Student PII: stored in isolated tables; never passed to LLM prompts; behavioral data uses anonymous student IDs

11. Non-Functional Requirements

Performance

Metric	Target
Course build — curriculum ready	< 14 min from brief submission (including Intelligence + Architect stages)
Course build — full 10-module course	< 4 hours end-to-end (parallel MediaForge stage)
Video render — 10-min lesson	< 7 min (FFmpeg on AWS Batch) p95
API response latency	< 200ms p50 / < 800ms p99 (excluding LLM time)
Marketplace search	< 500ms p95 (Algolia SLA)
Creator Studio page load	< 1.5s initial / < 600ms client-side navigation
Video stream start time	< 2s to first frame (CloudFront adaptive bitrate)
Quiz score calculation + feedback	< 250ms after submission
Discussion AI response	< 60s from question post to AI response
Course publish (idea → live URL)	< 90s from creator approval to indexed, purchasable URL

Throughput + Scalability

- API: AWS ECS Fargate auto-scaling; scale on CPU > 65%; target 1,000 concurrent API connections at MVP
- Pipeline: BullMQ job queue; up to 300 concurrent course build jobs; AWS Batch elastic scale for video rendering (no fixed capacity)
- Video CDN: CloudFront unlimited concurrent streams globally; S3 multi-region for low-latency origin
- Database: Supabase PostgreSQL with 1 primary + 1 read replica; PgBouncer connection pooling
- Targets: 10,000 MAU at launch; 100,000 MAU by Year 1; 1M MAU by Year 2

Reliability

Component	SLA
Creator OS + API	99.9% uptime (< 8.7 hrs/year)
LearnSpace (Student App)	99.95% (< 4.4 hrs/year) — direct revenue impact
Video CDN	99.99% — CloudFront SLA inherited
AI Pipeline	99.5% — failures queue + retry; partial results preferred over hard failure
Payment (Stripe)	99.99% — Stripe SLA inherited
Backup + Recovery	Hourly Supabase snapshots; S3 versioning; RPO < 1hr; RTO < 3hrs

Security

- Auth: Supabase magic link + Google/GitHub OAuth; optional TOTP 2FA; JWTs (1hr expiry) + rotating refresh tokens (30-day window)

- Authz: Supabase RLS on all tables; creators access only their own data; students access only enrolled courses
- API: HTTPS enforced; HSTS; CORS origin whitelist; input validation (Pydantic v2 on all endpoints); parameterized queries
- AI injection: user-provided content (topic brief, creator notes) sandboxed before LLM context injection; system prompt integrity validated per request
- Secrets: AWS Secrets Manager with rotation; API keys hashed (bcrypt); no secrets in env vars or logs
- Compliance roadmap: SOC 2 Type II within 18 months of launch

Cost Constraints

Cost item	Target
AI cost per full course build	< \$12 blended (Opus for scripts, Haiku for quizzes, ElevenLabs for voice)
AI cost per creator per month	< \$25 at 3 courses/month average usage (Creator tier)
Narration audio caching saving	35% ElevenLabs cost reduction via (script_hash + voice_id) cache
Infrastructure per active user/month	< \$0.80 (storage + CDN + DB)
AI as % of Creator tier revenue	< 15% at scale; declining with model efficiency improvements
Infrastructure at MVP	~\$200/mo at 500 creators; scaling to ~\$3K/mo at 15K creators

12. AI Safety and Governance

Bias Mitigation

- Scriptwriter and Launchpad system prompts include explicit diversity and inclusion instructions: gender-neutral language, diverse cultural examples, non-stereotyping in case studies
- Monthly bias audit: 5% random sample of generated content reviewed; findings published in quarterly transparency report
- Model-agnostic architecture — no single LLM lock-in; bias characteristics tracked per model version

Hallucination Handling

- [VERIFY] tags: all uncertain claims flagged inline; creator sees count in review UI; must explicitly dismiss each flag before video generation
- Source citation: Intelligence Agent output cites sources for all competitor data and market figures
- Confidence threshold: Optimizer recommendations include evidence count and confidence level; low-evidence recommendations labeled 'Hypothesis — needs verification'
- Failed verification → section flagged for creator review; never silently published

Content Moderation

- Topic intake screen: content policy check before pipeline starts (hate, misinformation, fraud, dangerous DIY, illegal). Hard reject with explanation; no resources consumed.
- Regulated domains: medical / legal / financial / pharmaceutical topics trigger mandatory disclaimer injection in lesson content and sales page
- Voice consent: mandatory consent confirmation with audit log before voice model training; ElevenLabs public-figure detection active
- Originality: all scripts through Originality.ai; > 12% similarity triggers regeneration; final score shown in Studio

Compliance

- GDPR: right-to-be-forgotten, data export, consent tracking, DPA with all sub-processors, data residency options for EU
- CCPA: California creator/student rights honored; no data sale; quarterly privacy policy review
- COPPA: age gate at registration; not designed for users under 13; enterprise clients deploying for younger learners confirm compliance independently
- SOC 2 Type II: target within 18 months of launch; audit preparation begins Month 9

Audit Logging

- Every pipeline run, agent decision, creator approval/rejection, student data access — immutable append-only log
- Retention: 7 years; exportable for regulatory review; searchable by creator / course / stage / date
- PII redacted from logs; pseudonymized user IDs in all AI traces; Langfuse stores prompt+response for 90 days

13. Monitoring and Observability

Application Monitoring

What	Tool + Alert
API latency + error rate	Datadog APM — P50/P95/P99 per endpoint; error rate; throughput. Alert: error rate > 1% for 5 min → PagerDuty
Application errors	Sentry — real-time exception capture; stack traces; release regression detection. New high-freq error → immediate Slack
Infrastructure health	Datadog Infrastructure — CPU, memory, disk per ECS task. Alert: CPU > 85% sustained → auto-scale trigger
Database	Supabase + Datadog — slow queries (> 300ms), connection pool, replication lag. Alert: lag > 30s → PagerDuty
Video render queue	BullMQ Dashboard — queue depth, failure rate, stuck jobs (> 20 min). Alert: depth > 200 → scale workers
CDN delivery	CloudFront real-time logs — cache hit rate, origin error rate, bandwidth. Alert: error rate > 0.5% → investigate origin
Uptime (external)	Datadog Synthetics — checks every 60s on 6 critical URLs. Alert: 2 consecutive failures → PagerDuty on-call
Payments	Stripe Dashboard — failed payments, dispute rate, payout failures. Alert: dispute rate > 0.5% → review fraud settings

AI-Specific Monitoring (Langfuse)

AI Metric	Tracking + Threshold
Prompt + response logs	Every LLM call: full prompt, response, token count, model, latency, cost per call. Searchable by course_id, agent_name, date
Pipeline stage latency	Per-stage P50 + P95 tracked daily. Alert: any stage P95 > 2× baseline
AI cost per course build	Cost attributed to course_id. Alert: single run cost > \$25
Content quality score	Weekly human review: 20 random courses rated 1–5 on accuracy, clarity, depth. Target avg > 4.2
Hallucination rate	Monthly SME audit on 5% sample. Target < 2%. Alert if monthly rate > 3.5%
Voice naturalness score	ElevenLabs quality metric + weekly blind panel (10 samples). Alert if drops below 3.5/5.0
Pipeline failure rate	% of builds completing without error. Target > 98%. Alert if < 96% in any 24h window
Model performance drift	Bi-weekly benchmark: 20 fixed test briefs. Alert if aggregate quality drops > 8% vs. baseline

Alerts Summary

- GMV drops > 35% vs. prior 7-day avg → Revenue + CEO Slack alert

- Daily AI spend > 150% of 7-day avg → Engineering + Finance alert
- Platform-wide completion rate drops below 40% → Product team review
- Refund rate > 8% in any 7-day window → Trust & Safety review
- Creator inactive > 45 days after first course publish → automated re-engagement email sequence

14. Testing Strategy

Unit Testing

- Pytest (Python backend services, agent logic, cost calculations, quiz scoring, certificate generation)
- Jest (Next.js components, API client, Stripe webhook handlers)
- Coverage target: 80% for core services, 60% for routes

Integration Testing

- Full pipeline E2E: 12 fixed topic briefs run through complete pipeline; validate output schema, script word count, video file existence, quiz question count, sales page sections
- API contract tests: all 30+ endpoints with mocked external APIs (ElevenLabs, Stripe, Algolia)
- Multi-tenant isolation: no cross-creator data leak; creator A cannot access creator B's courses
- Payment flow: full Stripe test-mode coverage (purchase, refund, failed card, 3DS, payout, affiliate commission)

AI Evaluation

Automated:

- Nightly pipeline tests on 12 fixed briefs; quality scored vs. established baseline; alert if drops > 8%
- Weekly: 25 randomly selected courses rated 1–5 by human reviewers (accuracy, clarity, pedagogical quality)
- Bi-weekly hallucination audit: 5% sample reviewed by domain expert panel

Human:

- Internal team dogfoods first — course quality ratings tracked in weekly scorecard
- 15 external alpha creators (Month 3)
- Closed beta with 100 creators (Month 4, pre-public launch)
- Quarterly red-team test: prompt injection, PII leakage, content policy bypass attempts

Load Testing

- k6: 300 concurrent pipeline builds + 5,000 concurrent video streams + 1,000 API req/sec
- Soak test: 48h sustained run — verify no memory leaks, no queue buildup
- Run monthly and pre-major release; verify all SLA targets hold under load

15. Deployment and DevOps

CI/CD Pipeline

- GitHub Actions on push to develop/main
- Stages: ruff + black (lint) → pytest → Jest → Snyk (security scan) → Docker build → ECR push → ECS deploy
- Auto-deploy to staging on merge to develop; manual promotion gate to production (Engineering Lead sign-off)

- Production deploy: blue/green via ECS service swap → automated smoke tests → 100% traffic shift → PagerDuty standby 1hr post-deploy

Environment Setup

Environment	Purpose	Notes
Development (local)	Local dev with Docker Compose	All external APIs mocked (VCR.py); local Supabase; Stripe test mode
Staging	Pre-prod testing; production mirror	Real AI APIs (rate-limited); test Stripe; 100 seeded courses + 500 students; E2E + load tests
Production	Live platform	AWS ECS Fargate + Vercel + Supabase + S3 + CloudFront; 24/7 monitoring; auto-scale active

Rollback Strategy

- Production rollback: one-command ECS service rollback to previous task definition; < 3 min estimated time
- DB migrations: additive-only for 2 releases; destructive migrations require dual-write period + explicit rollback script in PR
- AI model emergency rollback: model version pinned in config; revert to previous model in < 15 min via feature flag toggle; no deployment required
- Feature flags: PostHog — all new AI features behind flags; rollout: 1% → 5% → 25% → 100% with quality metrics checked at each threshold

16. UX and UI Requirements

Design Philosophy

Clean, functional, confidence-inspiring — not flashy. The AI's output quality is the product's differentiator, not the UI's visual complexity. Creators should feel like they have a brilliant, fast assistant. Learners should feel like they're in a focused learning environment with no distractions.

Core User Flows

Flow 1 — New Creator, First Course (< 30 min to first draft)

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Sign up → Voice Studio (optional, 10 min) → Topic Brief (2 min: topic + audience + depth)
→ Market Report screen (review + pick angle, 5 min) → Curriculum Review screen (reorder, 5 min)
→ Script Review (inline edit + [VERIFY] review, 10 min) → Video Preview (approve/regenerate)
→ Quiz Review (edit questions, 5 min) → Launch Setup (customize sales page + price, 10 min)

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→ Publish → live URL with Stripe checkout

Flow 2 — Daily Creator Operations

Dashboard → Revenue summary + new enrollments + pending improvement report
→ Analytics: review weekly AI insight narrative + drop-off heatmap
→ Discussions: review AI-escalated questions + post 2 verified answers
→ Revenue: check upcoming payout + affiliate performance

Flow 3 — Student Learning

Marketplace → Search/discover → Course detail (free preview 2 lessons) → Purchase
→ My Learning → Course Player → video + notes + Q&A → Module Quiz
→ Complete all modules → Capstone submission → Certificate generated → LinkedIn share

Creator OS — Key Screens

- Topic Brief: 3-field form (topic, audience, depth slider), language selector, estimated cost + time preview, 'Start Building' CTA
- Market Intelligence: demand score gauge, competitor grid (expandable), 3 angle cards (selectable), 'Proceed with this angle' CTA
- Curriculum Review: drag-drop module list, expand/collapse per module, inline lesson title edit, 'Approve Curriculum' CTA
- Script Review: per-lesson tabbed view (Script | Slides | Video), [VERIFY] flags highlighted in orange, 'Regenerate with note' field, word count + estimated delivery time
- Video Preview: in-browser player, chapter markers, 'Regenerate this video' with instruction field, approve/reject
- Quiz Editor: question list per module, answer distribution, difficulty tag, add/remove question, 'Looks Good' CTA
- Launch Setup: sales page preview (live HTML), headline A/B selector, price input with competitor tooltip, email sequence previews, 'Publish Course' CTA with launch checklist
- Creator Dashboard: revenue card + enrollments + completion rate + top course + improvement report banner + quick actions

LearnSpace — Student UI Screens

- Course Detail: hero, 'What you'll learn' bullets, curriculum preview (expandable), instructor bio, reviews, free preview CTA, buy CTA
- Course Player: video player, lesson sidebar with completion states, note panel (syncd to timestamp), Q&A tab, Resources tab (PDF workbook, slides)
- Quiz: progressive question display, progress bar, immediate feedback, score summary with module-level breakdown, retry option
- My Learning: enrolled courses + progress bars + next lesson shortcut + certificates earned
- Certificate: view + download PDF/PNG + LinkedIn share + verification URL + Open Badge export

Design System

- Aesthetic: clean, professional, trustworthy — blue accent (#2563EB) on white/light-gray surfaces; no dark patterns; no fake urgency
- Typography: Inter (web), system font (mobile)
- Components: Radix UI primitives (web) — accessible, keyboard-navigable
- Accessibility: WCAG 2.1 AA; captions on all videos; keyboard navigation for all critical flows; color contrast $\geq 4.5:1$

17. Timeline and Milestones

Phase-Wise Delivery Plan

Phase	Window	Deliverables
Phase 0 — Foundation	Month 1–2	AWS infra + Supabase schema + auth + BullMQ + LangGraph skeleton + Intelligence Agent + Architect Agent + Creator OS shell (brief + curriculum screens) + CI/CD + Langfuse
Phase 1 — Alpha Pipeline	Month 3–4	Scriptwriter + MediaForge (slides + voice + video) + Evaluator + Launchpad agents · Creator OS full review workflow · LearnSpace player + quizzes + certs · Stripe payments + creator payouts · invite-only alpha (30 creators)
Phase 2 — Public Beta	Month 5–6	Voice Studio · Optimizer Agent · Marketplace (Algolia) · Affiliate system · Analytics dashboard · White-label storefront (beta) · Public beta launch with waitlist — target 500 paying creators
Phase 3 — Growth	Month 7–8	Multi-language (6 languages) · Bulk enterprise batch · Advanced analytics (cohort, heatmap) · Promo code manager · Course repurposing engine · Adaptive learner paths · PWA mobile wrapper
Phase 4 — Enterprise	Month 9–12	SSO · SCORM/xAPI export · White-label full release · API public release · Compliance course templates · SLA tiers · SOC 2 audit begins · Enterprise ACV > \$30K target
Phase 5 — Ecosystem	Year 2	Native iOS + Android learner app · Live cohort classes · AI 1:1 tutoring · Creator collab (multi-instructor) · 20-language support · Blockchain credentials · \$24M ARR target

MVP Definition

"MVP done" = all 18 features (F1–F18) shipped and functional for the alpha cohort; Creator OS review flow complete; LearnSpace player + quizzes + certs live; Stripe payments processing; first paying student enrolled; landing page + waitlist live.

18. Business Model

Pricing Strategy

Tier	Price	Limits	Target
Starter	\$0/mo	3 course builds/mo · 5 modules max · EduGenie subdomain · marketplace listing · community support	Evaluators, students, hobbyists
Creator	\$45/mo (\$450/yr)	Unlimited builds · 18 modules · voice model (1) · 6 languages · custom domain · affiliate · full analytics	Coaches, consultants, educators
Studio	\$109/mo (\$1,090/yr)	Everything in Creator · 5 team seats · 8 parallel builds · white-label storefront · advanced analytics	Power creators, small agencies, L&D teams
Enterprise	Custom (\$600+/mo)	Unlimited everything · SSO · SCORM export · 30 parallel builds · SLA · dedicated CSM · API access	Corporate L&D, universities, platforms

Revenue Streams

Stream	Model + Projection
SaaS Subscriptions	Monthly/annual creator subscriptions — primary revenue. Target \$4M ARR by Month 18.
Course Sales Commission	5–12% platform take rate on course sales. Target \$30M GMV by Month 24 → \$1.5–3.6M platform revenue.
AI Build Credits	Starter users buy additional builds at \$8/build. Projected \$200K ARR by Month 18.
Enterprise Licensing	ACV \$30K–\$80K. Target 30 enterprise accounts by Month 18 → \$1.2M ARR.
White-Label Licensing	Agencies: \$500–\$2,000/mo per white-label instance. Target \$400K ARR by Month 24.

Path to MRR Milestones

Period	Key metrics
Month 3 — Alpha	30 alpha creators · 150 course builds · first paying students · \$5K MRR
Month 6 — Beta Launch	800 paying creators · \$35K MRR · 5,000 courses · \$800K GMV
Month 12	5,000 paying creators · \$180K MRR · 30,000 courses · \$12M GMV · 10 enterprise accounts

Period	Key metrics
Month 18	15,000 paying creators · \$500K MRR · 100,000 courses · \$35M GMV · 30 enterprise · \$6M ARR
Gross Margin (Year 1)	~62% — AI API costs high in early stage; AI as ~18% of Creator revenue
Gross Margin (Year 2)	~74% — better model routing, caching, volume discounts, infrastructure efficiencies
LTV:CAC Target	> 4:1 at 24-month LTV with 65% Year-1 creator retention

Cost vs Revenue

- COGS (AI inference): target < 15% of revenue at scale (model routing + caching)
- Infrastructure: ~\$200/mo MVP → ~\$3K/mo at 15K creators → ~\$20K/mo at 100K creators
- Payback period: < 14 months CAC payback at \$45 ARPU and estimated \$500 blended CAC

19. Risks and Assumptions

Risk Register

Risk	Likelihood / Impact	Mitigation
AI content quality fails creator expectations — they don't trust it enough to publish	High / Critical	30-creator alpha with direct feedback loops before beta; full edit + regeneration at every stage; position as '80% first draft'; monthly SME audits with public quality report
ElevenLabs raises prices or degrades quality at scale	Medium / High	Abstract integration layer; 2 tested fallback voice providers (Cartesia, Resemble) ready to activate; in-house TTS roadmap for Year 2
Big Tech (Adobe, Google, Canva) launches competing AI course creation tool	High / High	Move fast on category definition; build creator community moat before Year 1; data flywheel (completion benchmarks, optimization intelligence) improves with scale — pure feature parity won't replicate
Marketplace flooded with low-quality AI courses, damaging platform reputation	High / High	Quality gate before marketplace listing (schema validation + random human spot-check); 'EduGenie Verified' badge; creator reputation scoring; content moderation team from Month 6
High creator churn: creators build one course and leave — low LTV	High / High	Optimizer reports drive re-engagement; course update feature creates ongoing value; 'second course' nudge at 30 days post-publish; show revenue potential of additional courses in audience segment
AI inference costs grow faster than revenue	Medium / High	Model routing (Haiku for low-complexity tasks); ElevenLabs narration caching (35% cost reduction); volume pricing negotiations; cost dashboard per creator tier
Voice cloning misused (impersonation, non-consensual)	Low / Critical	Mandatory consent with audit log; ElevenLabs public-figure detection; voice model locked to account; DMCA process; immediate suspension for confirmed violations
Student learning outcomes poor — courses don't actually teach well	Medium / Critical	Bloom's taxonomy rigor in Architect agent; completion + quiz rates tracked per course with alerts; independent learning outcome study at Month 12; Optimizer Agent continuous improvement loop

Key Assumptions

- Creators accept AI-generated content as a strong starting point — they value speed over perfection and will refine, not reject
- The 4-hour full course build vs. 150–300 hours manually is a compelling enough value prop for \$45–\$109/month subscription
- ElevenLabs voice quality at Creator tier is acceptable to online course learners — blind test data needed in alpha
- AI-generated courses will achieve comparable marketplace ratings to hand-crafted courses when creator reviews before publishing
- Platform take rate of 5–12% is acceptable given the value of AI creation + marketplace distribution + ongoing analytics

- Corporate L&D will accept AI-generated training content subject to SME review — enterprise pilots in Month 7–9 will validate

20. Open Questions

Question	Owner / Target date
What is the minimum acceptable voice clone quality (naturalness score) for online learners? We need blind test data from alpha creators and their first student cohorts.	AI Engineering + UX Research — Month 2
Should EduGenie OS offer a marketplace where creators pay domain experts to review AI-generated content as a trust-building service?	Product + Business Dev — Month 3
Is the 5% platform take rate model or a 0% take rate at a higher subscription price more attractive to creators? Should we offer both and let creators choose?	CEO + Revenue — Month 3
How do we handle courses on rapidly-evolving topics (AI, regulatory changes)? Do we add a 'freshness score' and proactive update reminders?	Product + AI Safety — Month 2
Should creators be able to upload their existing course content (video, slides, PDF) as input and have EduGenie restructure + improve it?	Engineering + Product — Month 3
What is the right minimum quality threshold for marketplace listing? All completed courses, or only those meeting a quality floor?	Product + Trust & Safety — Month 4
Should we seed the marketplace with EduGenie-built courses on popular topics before creator onboarding to solve the chicken-and-egg problem?	Product + Growth — Month 2
Should AI discussion board responses disclose to students that they are AI-generated, or should creators have the option to present them as their own?	Product + Legal — Month 2
What enterprise LMS integrations are required for the first enterprise contract — Workday Learning, SAP	Enterprise Sales + Engineering — Month 6

Question	Owner / Target date
SuccessFactors, Cornerstone? What is the minimum viable set?	
Should EduGenie invest in fine-tuning a proprietary instructional design model on high-completion course data, or remain on foundation models?	AI Engineering + CTO — Month 4

— End of Document — EduGenie OS PRD v1.0 | May 2026 | CONFIDENTIAL